

TECHNISCHE UNIVERSITÄT BERLIN

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## Investigating generative capacities of LLMs for emerging disinformation detection

A Proactive Framework for Flagging Disinformation Posts via Semantic Similarity

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MASTER THESIS

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# 1 Motivation

The spread of false information poses significant and increasingly relevant challenges today. Fake news has been shown to have directly influenced a wide range of domains: from politics, such as presidential elections (1, 2), to healthcare, as observed during the COVID-19 pandemic and its resulting *infodemic* (3), to real world events, with cases such as the "Pizzagate" conspiracy that ended in a real shooting (4). With the rise of the Internet and social media, the impacts have become more pronounced, as disinformation now spreads faster and more widely than before. Social media platforms such as Twitter, Facebook and Reddit, provide ideal environments for rapid communication exchange, often with little verification of sources and minimal regulation. In fact, studies show that fake news is spread six times faster than real news on Twitter (5). Moreover, algorithmic personalization (e.g. through recommendation systems) tends to reinforce users' existing beliefs and isolate them from opposing perspectives, creating so-called *filter bubbles*. These dynamics amplify the reach of misleading content and highlight the urgent need for detection and mitigation of disinformation.

Existing solutions to this problem focus on *debunking* false posts by correcting misinformation and mitigating its spread. This can occur in more traditional ways - for instance, through fact-checking websites that rely on expert investigation of news content - or in automated ways. The latter ranges from classical machine learning approaches that classify if a piece of news is authentic or fabricated, to more recent transformer-based architectures. State-of-the-art methods for automated fake-news detection often include encoder-based architectures like BERT and decoder only Large Language Models (LLMs) such as GPT, frequently in combination with external knowledge bases or social context (6).

However, little attention has been paid to the generative capabilities of LLMs as a tool to identify fake news. Instead of waiting for fake news to appear and then labeling it, a more proactive approach would be to use LLMs to generate synthetic disinformation based on real events and then compare that to real claims circulating online. Analyzing the similarity between synthetic and organic false information can provide valuable insights into the check-worthiness of real claims. This thesis therefore investigates whether generative models can be used as a proxy/an approximation for real world fakes and be leveraged to assess if a social media claim needs fact-checking before it is widely spread. The main research question then is whether LLMs can be employed not only as classifiers but also as **controlled generators of fake content, and whether the resulted fakes can help identify which claims are most likely to be false.**

This exposé is organized as follows: Section 2 reviews related work in fake news detection, transformer-based classification and generative approaches. Section 3 outlines the goals and expected contributions of this thesis. Section 4 presents the suggested methodology for experimenting with the generative capabilities of LLMs. Finally, Section 5 lays out the planned schedule for finishing this work.

## 2 Related Work/Literature Review

### 2.1 Fake News: Definitions and Characteristics

The terminology around "fake news" is often used imprecisely and interchangeably, with related concepts such as misinformation, disinformation, rumor, among others. However, it is

important to have precise definitions to avoid ambiguity and to formulate appropriate mitigation strategies.

According to the Council of Europe’s Information Disorder Report (7), there are three important concepts in this context: mis-, dis- and mal-information. **Misinformation** refers to false information shared, but with no intent to harm. One example would be a satirical article that is mistakenly taken seriously, or a post with an inaccurate translation. **Disinformation**, on the other hand, is defined as false information deliberately created and shared with the intention to cause harm. This can range from misleading headlines and manipulated content to entirely fabricated content. Although the term "fake news" is the most widely-used term for it, it is vague and misleading within the research community, and therefore will be avoided. Finally, **mal-information** refers to real information shared with the intent to cause harm, such as a data leak of private information.

The deceptive nature of disinformation leads to a linguistic style notably different from authentic content. For instance, articles from credible news sources typically have a more balanced and objective tone and are supported by fact-checked sources. Fake news, on the other hand, is characterized by sensationalism, biased language and a lack of credible sources (8).

Understanding the context where information, and therefore disinformation, is consumed is essential for formulating mitigation strategies. Social media has become an important platform for that. According to a survey conducted by the Pew Research Center in 2025, 53% of U.S. adults said they used social media for accessing news "often" or "sometimes", with Facebook and Youtube leading as sources (9). Even though other platforms have a smaller audience, they are mainly consumed as news source - e.g. 57% of X (formerly Twitter) users get their news there (10).

## 2.2 Traditional Approaches to Fake News Detection

A fundamental way to combat false information is through traditional human fact-checking. This involves trained journalists and specialists manually verifying claims by cross-referencing a piece of news with primary sources, such as official reports, academic studies, and other credible documents. This is done differently according to the context, with labels varying from simple binary  $\{true, false\}$  to a graded scale of truthfulness. One well-known example is *PolitiFact*, a fact-checking journalism website established in 2007, which classifies claims with labels  $\{true, mostly\ true, half\ true, mostly\ false, false, pants\ on\ fire\}$  (11). Other classification methods include  $\{true, false, unverified\}$ (12).

While human fact-checking serves as a foundation, it is time-consuming and highly resource-intensive, and does not scale well given the speed at which disinformation spreads online. A social study analyzing the spread of claims on Twitter over an 11-year period concluded that "falsehood diffused significantly farther, faster, deeper, and more broadly than the truth in all categories of information" (13). In response to these limitations, **automated fake news detection techniques** were developed. Early methods included classic machine learning techniques to identify whether a news instance is fake. One example is a Support Vector Machine Classifier that uses linguistic features - such as n-grams or punctuation - to reach a decision(14). Other examples include Naive Bayes Classifier, Random Forest, among others (15, 16). More recent approaches leverage deep learning models, such as Convolutional Neural Networks (CNNs) or Long Short Term Memory (LSTMs), capable of capturing semantic meaning in news content (17).

These approaches, while foundational, are limited by their reliance on shallow, hand-engineered features and often fail to capture more complex contextual and semantic cues.

## 2.3 Transformer-based approaches for disinformation detection

Since the introduction of the Transformer architecture, these models have become the dominant approach in the field of Natural Language Processing (NLP), shifting the landscape of automated fake news detection. Their ability to capture semantic relationships make them a preferred method for classifying disinformation (18). This section discusses transformer-based architectures for this task, focusing first on its classification and then on their generative capabilities.

### 2.3.1 Transformer-based models as classifiers

Many transformer-based solutions rely on encoder-only models for fake news classification. These approaches typically involve fine-tuning BERT-like architectures such as *DistilBERT* (19) or *RoBERTa* to distinguish between real vs. fake news based on contextual and linguistic cues. Such models have shown reliable performance and high accuracy in identifying disinformation (20).

Large Language Models also introduce advantages in fake news classification tasks because of their strong semantic understanding, reasoning ability and multimodal capabilities - the latter particularly useful in the context of social media, where text is often accompanied by images or videos (18). Unlike traditional classifiers, these models can reason over the content, draw inferences and justify their conclusions. Through few-shot prompting, for example, an LLM can be guided to evaluate different aspects of a claim (e.g. emotional tone, internal consistency, etc.) before deciding whether it is true or false. Moreover, they have potential for explainability, because the model’s rationale can be shown to a human fact-checker or moderator, to support or oppose the automated classification (21).

However, LLMs are primarily designed for text generation rather than classification, which results in higher computational costs and more latency. Running large decoder-only models such as GPT-style architectures typically requires significant GPU memory and parallel infrastructure, making it more costly to deploy at scale on down-stream tasks. Their longer inference time is also problematic in high-speed information environments like social media, where delays in detecting false claims increases the potential harm caused by disinformation.

Comparative research supports these findings. One study compared the performance of encoder- (BERT-based) vs. decoder-only (LLMs) language models for claim verification on social media. It was concluded that fine-tuned BERT models achieved lower computational costs and lower latency for classifying, while maintaining comparable accuracy (22). There are studies that explore a hybrid approach, using the LLM generated label as a "pseudo-label" (a weak classification) and combining it with the prediction of a fine-tuned BERT based model as main label (8).

### 2.3.2 LLMs as generators of disinformation

Beyond classification, prior work has explored Large Language Models as disinformation generators and confirmed its efficiency producing deceptive content. One early study introduced a generative model *GROVER*, which writes false news articles based on a given headline and

metadata. The results show that **machine-generated propaganda was often considered trustworthy** by human evaluators, indicating its potential to mislead readers (23). Another research article comparing human- and machine generated tweets concluded that **synthetic content not only deceived evaluators better**, but also required less time and effort to interpret (24).

Expanding on this, several works focused on systematically generating synthetic fake news datasets, which offers valuable insights into effective prompting strategies and model architectures for disinformation. One example is the *MegaFake dataset* (25), which systematically produces various fake news styles based on real events. It uses different prompting strategies to generate four types of disinformation categories, of which two are interesting for this thesis: *{style based fakes}* (rewriting an article with sensational or emotionally triggering tone) and *{content based fakes}* (manipulate existing news by modifying attributes).

Another study has employed LLMs as both generators and detectors of disinformation articles based on real events (26). On the generation side, they adopted a perturbation-based approach with different levels of modification: *minor, major and critical*. Each generated instance included three components - the content C, an impersonator R, and an instructor I - adopting a persona-based prompting. They also included a subsequent filtering step that consisted of checking the generated outputs for hallucinations and semantic alignment to the original articles, ensuring that only valid generated data was retained. The findings revealed that persona-based prompting effectively bypassed most model safety mechanisms, enabling the generation of malicious or misleading text. Moreover, approximately 38% of the generated samples exhibited hallucinated or misaligned content, underscoring the importance of validating generation quality and factual alignment before proceeding to the detection phase.

Recent work, such as "Disinformation Capabilities of large Language Models" (27), compares different model families when prompted with false headlines. It shows that modern ones (e.g. GPT-4 and Mistral) produce highly realistic fake news and are steerable through prompting, typically without triggering safety mechanisms or refusal messages. Furthermore, it shows that larger models generate articles more similar to real news, and that adding more context (for example, headline and abstract), improves the level of alignment between prompt and text.

In conclusion, previous work confirms the **high efficacy of LLMs in producing realistic disinformation instances** and they provide valuable insights into different prompting strategies, choice of model and data, that inform the design of this work's experimental pipeline.

### 3 Goals

This thesis is an exploratory project with the primary goal of investigating if Large Language Models can be leveraged **to generate disinformation content that is useful for identifying new instances of false or misleading social media claims**. The work integrates two objectives: understanding the generative capabilities of LLMs in creating realistic fake content and evaluating the usefulness of these synthetic fakes as proxies for proactive disinformation detection. The research is guided by the following questions:

- RQ1: Can LLMs be prompted to generate disinformation content that follows specific narratives or stylistic biases?

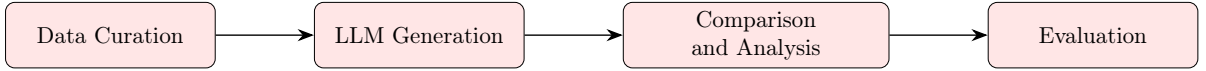
- Sub-Q1.1: Which prompting strategies are most efficient at producing synthetic fakes that are semantically similar to human-authored ones?
- Sub-Q1.2: How do characteristics of generated fakes differ across various LLM architectures and model families (e.g. open-source vs. proprietary, censored vs. uncensored and models originated from different geopolitical contexts)?
- RQ2: **Can synthetic fakes serve as proxies for identifying false social media posts?**

The main scientific contribution of this thesis is the development of a proactive, automated technique for assessing the likelihood that a social media claim is false. By being fully automated, this method scales well and overcomes the time limitations of human fact checking. Furthermore, this approach is proactive: instead of waiting for rumors to be posted and then refuted, a system could automatically generate synthetic fakes as soon as a relevant news article is published. It could then compare freshly published social media posts for semantic similarity with the generated fakes, flagging highly similar entries for further review.

The expected contributions of this thesis are twofold. First, to demonstrate that LLMs can be successfully employed to generate fake social media claims that are highly similar to human authored ones. This includes an evaluation of different models and prompting strategies to identify the most effective methods for creating a false claim proxy. Second, to show that similarity search between machine and human generated posts can serve as a practical and proactive technique for flagging possible disinformation posts.

## 4 Methodology

The methodology consists of four main parts and is structured as follows.



### 4.1 Data Curation

The data curation consists of collecting a series of real news articles of recent events  $R = \{R_1, R_2, \dots, R_n\}$ . For each entry in  $R$ , a set of human-authored fake and real claims about the same real-world event made on social media is gathered, with  $H_{fake}, H_{real} = \{h_{i1}, h_{i2}, \dots, h_{ik}\}$ , each accompanied by a veracity label  $y_{h_{ik}} \in \{0, 1\}$ , where 0 means factual (true) post and 1 disinformation post. Metadata such as timestamps or engagement indicators may also be included.

This curated data is partitioned in two sets to serve different experimental goals:

1. **Core validation set:** This set is necessary for experimental setup and methodology validation. Its purpose is to select the optimal components for the pipeline, including prompting techniques, LLM variants, embedding models, similarity thresholds and evaluation metrics. It will be derived from **existing, established datasets** that link a verified news article to multiple social media posts. This should provide a controlled environment to test the assumptions, debug the pipeline and make design-related decisions such as prompt-tuning. Candidate datasets include *{FakeNewsNet (28), PHEME (12), Fakeddit (29)}*. For more information on the dataset characteristics and eventual problems, check the tables in the Appendix A.

2. **Generalization test set:** This set is designed to test the real-world applicability and generalization capabilities of the methodology against very recent news events **published after a cut-off date (DD.MM.2025)**. These will be very up-to-date news and can be confidently assumed to be outside the training corpus of the proposed LLMs. This set will be significantly smaller and fully curated at the Quality & Usability Lab at the TU Berlin, following a **hybrid automated and human-validated fact-checking** workflow, leveraging an existing automated fact-checker. The construction of the dataset will consist as follows:

- (a) Collect a series of recent social media claims referring to each selected event using the Exorde Labs dataset from the XplaiNLP group
- (b) Use the automated fact-checker (by Sahitaj et al. 2025) to make sure the claims  $H$  match the selected event  $R$
- (c) Select approximately 150 claims with the highest match scores and get their veracity labels  $y_{h_{ik}}$
- (d) Perform manual validation to confirm that the paired  $(r_i, h_{ik})$  are a match

This dataset represents a realistic, noisy and very recent distribution of user-generated content and it's used for out-of-distribution evaluation.

The exact datasets and news topics are yet to be determined, but there are some aspects that need to be considered for having a controlled and comparable environment across both settings. For instance, all social media posts will originate from the same platform, be written in English, have a pre-defined character-length range, and have the same metadata (e.g. timestamps). By the end of this process, there should be **pairs of human-made social media claims - both real and fake - and a corresponding news post:**  $(r_i, H_{i,real}, H_{i,fake})$ .

## 4.2 LLM Generation

The LLM generation phase is where, for each selected news event  $R_i$ , several disinformation posts  $S$  are generated using different *language models*  $M$  and *prompting strategies*  $P$ . The resulting synthetic dataset can be represented as  $S_i = \{s_{i,m,p} | m \in M, p \in P\}$ , where  $s_{i,m,p}$  is a generated claim for a given real news event  $R_i$ .

The exact parameter choice will be determined later, but the following aspects are to be taken into account:

- **Language models  $M$ :** A diverse set of LLMs will be tested, including both open source and proprietary, uncensored and censored models, as well as models originating from different geopolitical contexts (e.g. European, US-American and Chinese). Candidate models include  $\{LlaMa, Mistral, GPT-4, Qwen, Falcon\}$ .
- **Prompting strategies  $P$ :** Different techniques will be used to generate distinct forms of disinformation, such as an emotionally charged or sensationalist tone, and/or a specific political framing (left/right). Each prompt defines a controlled condition in which the LLM is asked to rewrite a given news event following a prescribed narrative bias. Zero- and few-shot, persona-based and adversarial prompting will be considered.

The exact prompt-engineering strategy will be refined during the experimental phase. However, in order to generate false yet realistic and human-like posts, several design choices are taken into consideration:

- **Chain-of-thought prompting:** The goal is to construct synthetic fakes not arbitrarily, but in a way that explicitly considers context and target audience. This can be achieved by guiding the model through intermediate reasoning steps. For instance, given a news event, the model could first extract the political context (e.g., identify the main stakeholders, controversies, etc.), then infer the main political positions that are associated to such an event, and only afterwards be instructed to rewrite the news in a way that reinforces a specific, potentially misleading claim aligned with one of these positions.
- **Few-shot prompting:** Prior work in this area often relies on zero- or one-shot prompting, which does not explore the potential of showing examples and giving more context to the LLM to produce its response. By using few-shot prompting, the model can “learn” from a small set of curated examples that contain typical stylistic and linguistic features of social media posts (e.g., use of slang, emojis, hashtags, emotionally charged language). This may help the model produce disinformation proxies that are more similar to actual user-generated content. One possible issue with this approach is that the model may overfit to the provided examples and replicate their wording too closely. This risk will be taken into account and the prompts will be designed to minimize such imitation.
- **Post-processing plausibility checks:** As a post-processing step, the generated posts could be filtered to ensure both relevance and plausibility. First, to reduce the risk of pure hallucinations (i.e., content that is almost entirely unrelated to the original news event), an automatic alignment or similarity metric between the synthetic fake  $S$  and the corresponding news event  $R$  could be applied, discarding posts that deviate too far from the source context. Second, additional criteria may be used to select disinformation posts that are false but still believable, for example by filtering out texts that are overly incoherent or implausible.

### 4.3 Comparison and Analysis

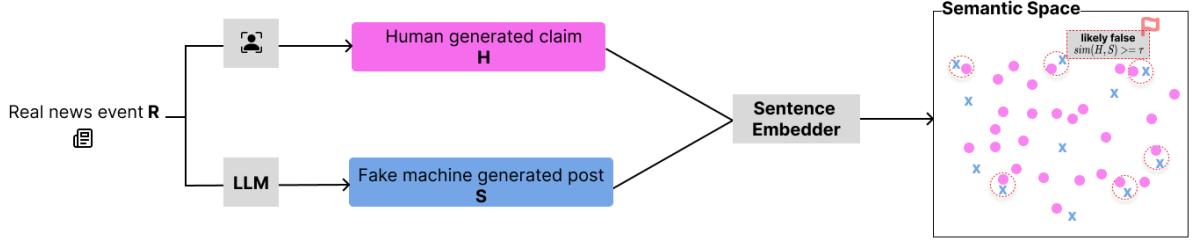
In this stage, semantic similarity between synthetic and human-authored posts is evaluated in order to identify which human posts are most likely to be false or misleading. The comparison consists of embedding human generated fake and real claims  $H_{real}, H_{fake}$  as well as synthetic fakes  $S$  in the same semantic space using a sentence embedding model such as  $\{S\text{-BERT}, GTE, MPNET, EmbeddingGemma\}$ . This results in an embedding space where entries close to each other have high semantic-textual similarity.

From there, the claims can be compared using similarity metrics such as *cosine similarity*. Other metrics, such as dot-product or angular-similarity, will also be considered in the process. As a comparable method, the *BM25* ranking algorithm will be used to perform a lexical similarity scoring, allowing comparison with the proposed embedding-based approach. Based on these scores, an approximate nearest neighbour search (k-ANN) can be performed to retrieve the top  $k$  most semantically similar human-authored posts for each synthetic fake, forming the set  $N_K(s) \subset H$ . Alternatively, a threshold-based approach can be applied, in which human posts with similarity values above a predefined cutoff are automatically flagged as potential fakes. The results serve as a claim falsity indicator, where **posts exhibiting high semantic proximity to synthetic disinformation are flagged as highly likely to be false**.

#### 4.4 Evaluation

The final phase is a general evaluation of the results, using both quantitative and qualitative analysis. The quantitative metrics include values such as *precision and recall at  $k$* , which show how many of the flagged claims  $H$  are real fakes. Other more general conclusions are to be drawn, e.g. which model  $M$  produced the most similar synthetic fakes, or which prompting strategy  $P$  was the most efficient.

An overview of the proposed pipeline can be seen in the following diagram:

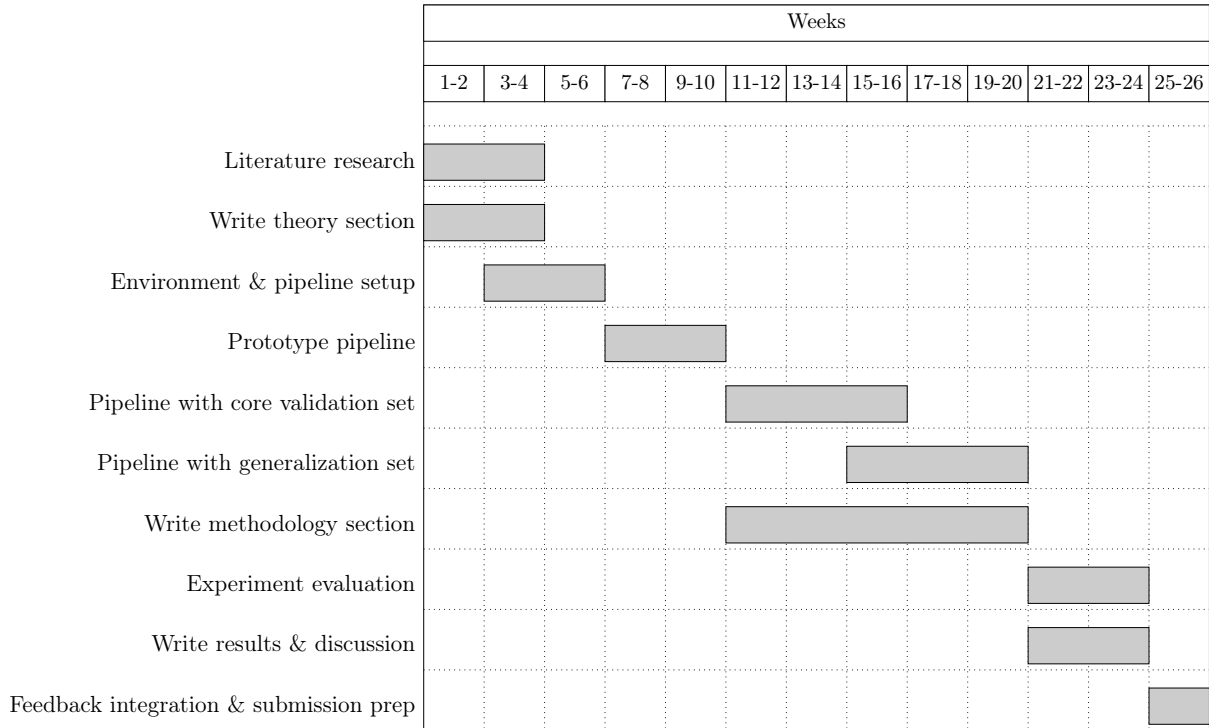


**Figure 1:** Overview of the proposed pipeline

### 5 Time Planning/Thesis Structure

The thesis will be supervised between MTEC lab and XplainNLP group at faculty IV of the Technische Universität Berlin, as well as Robert Nickell at Bucknell University. A biweekly overview of the schedule can be seen here:

The planned timeline is structured as follows. During the first four weeks, the primary focus will be on literary research, reviewing previous experiments in the field, with particular attention to the benchmarks that have been employed, the datasets used, and their respective limitations. The following six weeks will focus on setting up the work environment and implementing a preliminary version of the proposed pipeline. Subsequently, initial tests will be conducted using an existing dataset (the core validation set), followed by experiments on a generalization consisting of newly collected social media instances. The next month will be about evaluating the experimental results, analyzing encountered challenges and provided insights. Finally, the last month will be dedicated to the formal writing of results and discussions, as well as completing the thesis.



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## A Dataset Comparison for "Core Validation Set" Candidates

To support a more informed selection of the dataset for the "core validation set" in the experiments, a summary table was created. It is split into two parts for layout purposes and presents the datasets’ properties, access methods, and key constraints. All these aspects will be addressed once the official thesis period begins, alongside potential solutions to the key problems identified.

**Table A: Dataset Properties**

**Table 1:** Datasets: Focus, Size, and Labels (Part 1/2)

| Dataset      | Platform           | Modality                      | Domains                | # Posts/news | Labels                          |
|--------------|--------------------|-------------------------------|------------------------|--------------|---------------------------------|
| FakeNewsNet  | Newspapers/Twitter | News articles shared          | Politics/Entertainment | 65k          | real/fake                       |
| <b>PHEME</b> | <b>Twitter</b>     | <b>Rumors</b>                 | <b>Various</b>         | <b>62k</b>   | <b>rumor/non-rumor (T/F)</b>    |
| NELA-GT-2020 | Newspapers         | News articles based on Tweets | Various                | 1.7M         | unreliable, mixed, reliable     |
| Twitter15/16 | Twitter            | Claims                        | Various                | 2k           | rumor/non-rumor (T/F)           |
| LIAR         | Various            | Claims                        | Politics               | 12k          | Politifact’s labels (6-way)     |
| Fakeddit     | Reddit             | Multimodal (64% text+image)   | Various                | 1M+          | 2/3/6-way classification        |
| CREDBANK     | Twitter            | Event credibility             | Various                | 60M          | accurate, inaccurate, uncertain |

**Table B: Access and Constraints**

**Table 2:** Datasets: Access, Time Period, and Problems (Part 2/2)

| Dataset      | News Article? | Access Method      | Retrieval Type           | Key Problems                                              |
|--------------|---------------|--------------------|--------------------------|-----------------------------------------------------------|
| FakeNewsNet  | Yes           | Twitter API        | Only IDs                 | Costly/hard access (API); Only IDs.                       |
| <b>PHEME</b> | <b>No</b>     | <b>Huggingface</b> | <b>Posts retrievable</b> | <b>No associated news articles.</b>                       |
| NELA-GT-2020 | Yes           | SQLite DB          | Posts retrievable        | Veracity labels on news articles, not tweets.             |
| Twitter15/16 | No            | Huggingface        | Posts retrievable        | No associated news articles; Very small size.             |
| LIAR         | No            | Huggingface        | Claims retrievable       | Too many input formats (needs filtering).                 |
| Fakeddit     | No            | Uncertain          | Claims retrievable       | Multimodal (needs text-only filtering); Access uncertain. |
| CREDBANK     | Yes           | Twitter API        | Only IDs                 | Costly/hard access (API); Old data; Only IDs.             |